Note 2 - Types of Functions

The Types of Functions

Understanding function types and their rules can help us in finding their domains.

- **Linear**:
$$y = mx + b$$
In the above equation, m is the slope and b is the y-intercept.

- **Polynomial**:
$$f(x) = ax^n + bx^{n-1} + \cdots + cx^2 + dx + e$$
where the coefficients and constant are real numbers, or $a, b, c, d, e \in \Re$. The exponent, n, is the degree of the function. Common types of polynomials are:
  - Constant $n = 0$
  - Linear $n = 1$
  - Quadratic $n = 2$
  - Cubic $n = 3$

- **Power**:
$$f(x) = kx^a$$
where $k, a \in \Re$ and $k, a \neq 0$. Essentially, a single-term polynomial function where a variable base is raised to fixed exponent.

- **Rational**:
$$f(x) = \frac{P(x)}{Q(x)}$$
where $P(x)$ and $Q(x)$ are polynomial functions.

- **Algebraic**:
$$f(x) = ax^n + bx^{n-1} + \cdots + cx^2 + dx + e$$
The set of all real numbers (inputs), x, for which the function is defined or exists.

- **Trigonometric**:
$$f(x) = ax^n + bx^{n-1} + \cdots + cx^2 + dx + e$$
The set of all real numbers (inputs), x, for which the function is defined or exists.

- **Exponential**:
$$f(x) = b^x$$
The set of all real numbers (inputs), x, for which the function is defined or exists.

- **Logarithmic**:
$$f(x) = \log_b x$$
The set of all real numbers (inputs), x, for which the function is defined or exists.